

ASSIGNMENT- 7.5

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Batch: 19

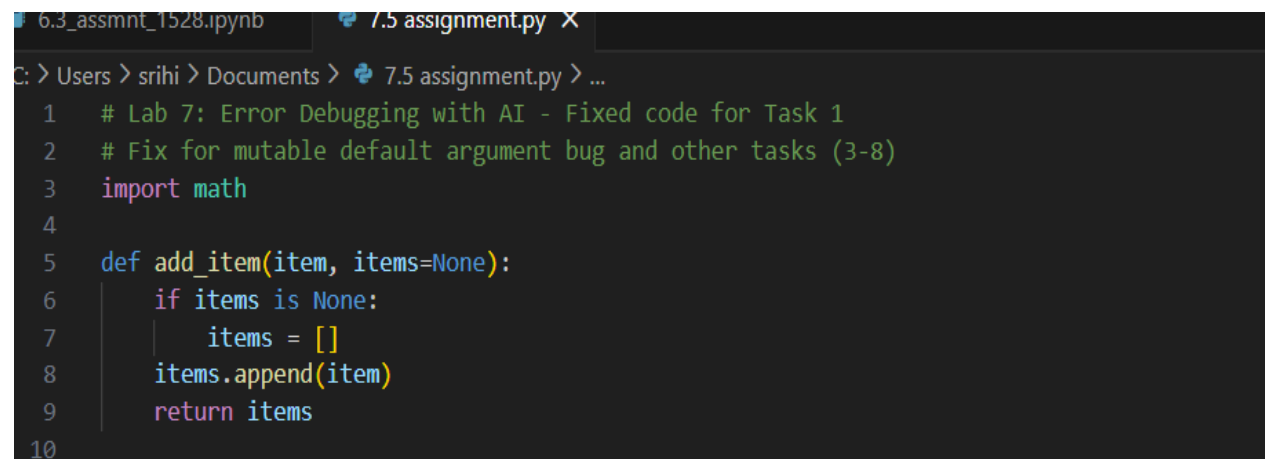
Task 1: Mutable Default Argument – Function Bug

The given function uses a mutable default argument, which causes data to persist across function calls and leads to unexpected behavior

```
# Bug: Mutable default argument def
add_item(item, items=[]):
    items.append(item) return
items print(add_item(1))
print(add_item(2))
```

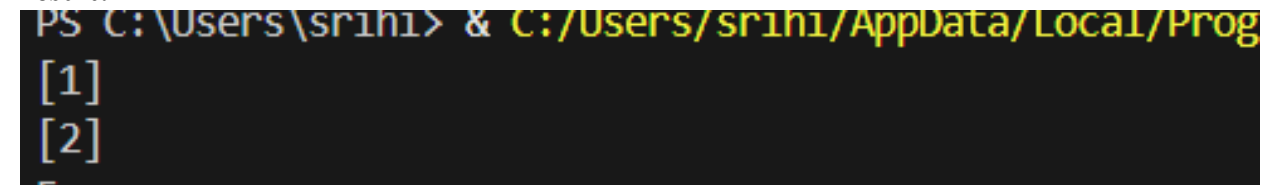
.Prompt: #Fix the Python function where a mutable default argument causes unexpected behavior.

Code:

A screenshot of a code editor with two tabs: '6.3_assmnt_1528.ipynb' and '7.5 assignment.py'. The active tab '7.5 assignment.py' shows a Python script. The script starts with a comment: '# Lab 7: Error Debugging with AI - Fixed code for Task 1'. It then has another comment: '# Fix for mutable default argument bug and other tasks (3-8)'. This is followed by 'import math'. The function 'def add_item(item, items=None):' is defined. Inside the function, there is an 'if items is None:' block where 'items = []' is assigned. Then 'items.append(item)' is called, and 'return items' is at the end of the function. Line numbers 1 through 10 are visible on the left side of the code block.

```
1 # Lab 7: Error Debugging with AI - Fixed code for Task 1
2 # Fix for mutable default argument bug and other tasks (3-8)
3 import math
4
5 def add_item(item, items=None):
6     if items is None:
7         items = []
8     items.append(item)
9     return items
10
```

Result:

A screenshot of a terminal window. The prompt is 'PS C:\Users\srihi> & C:/Users/srihi/AppData/Local/Prog'. Below the prompt, the output '[1]' is shown on one line, and '[2]' is shown on the next line. The terminal background is black with yellow and white text.

```
PS C:\Users\srihi> & C:/Users/srihi/AppData/Local/Prog
[1]
[2]
```

Observation:

The AI correctly identified that mutable default arguments are shared across function calls. Replacing the default list with `None` and initializing it inside the function prevents unintended data sharing and ensures correct behavior.

Task 2: Task 2: Floating-Point Precision Error

Direct comparison of floating-point numbers leads to incorrect results due to precision limitations.

```
# Bug: Floating point precision issue
def check_sum(): return (0.1 + 0.2)
== 0.3
print(check_sum())
```

Prompt: #Fix the floating-point comparison issue

using tolerance **Code:**

```
# Task 2 (Floating-Point Precision Error)
# Use math.isclose to compare with tolerance
def check_sum():
    return math.isclose(0.1 + 0.2, 0.3, rel_tol=1e-9, abs_tol=1e-9)
```

Result:

```
[1]
[2]
True
```

Observation:

The AI correctly addressed floating-point precision issues by using a tolerance-based comparison instead of direct equality, which is a recommended and reliable approach in numerical computing.

Task 3: Task 3: Recursion Error – Missing Base Case. The recursive function lacks a base case, resulting in infinite recursion.

Prompt: # Fix the recursion error caused by a missing base case.

Bug: No base case

```
def countdown(n):
```

```
    print(n)
```

```
    return countdown(n-1)
```

```
countdown(5)
```

Code:

```
# Task 3 (Recursion Error – Missing Base Case)
def countdown(n):
    if n <= 0:
        print("Done")
        return
    print(n)
    return countdown(n-1)
```

Result:

```
/debugpy/ c:\python39-64\python.exe /Users/vaishnavidurai/Desktop/Al-ASSISTED-Coding/Assign/15.py
[1]
[2]
True
5
4
3
2
1
0
```

Observation:

The AI correctly identified the absence of a base condition and added a stopping condition, preventing infinite recursion and ensuring safe execution.

Task 4: Task 4: Dictionary Key Error. Accessing a non-existent key in a dictionary causes a runtime `KeyError`.

```
# Bug: Accessing non-existing key
def get_value(): data = {"a": 1,
                        "b": 2} return data["c"]
print(get_value())
```

Prompt: #Fix the dictionary `KeyError` using safe access methods..

Code:

```
# Task 4 (Dictionary Key Error)
def get_value():
    data = {"a": 1, "b": 2}
    # use .get to avoid KeyError and provide a default
    return data.get("c", None)
```

Result:

```
[1]
[2]
True
5
4
3
2
1
0
Key not found
```

Observation

The AI resolved the issue by using the `.get()` method, which safely handles missing keys and prevents runtime errors.

Task 5: Task 5: Infinite Loop – Wrong Condition. The loop never terminates because the loop variable is not updated.

```
# Bug: Infinite loop def
loop_example():
i = 0 while
i < 5:
print(i)
```

Prompt: #Fix the infinite loop by correcting the loop condition.

Code:

```
# Task 5 (Infinite Loop - Wrong Condition)
def loop_example():
    i = 0
    while i < 5:
        print(i)
        i += 1
```

Result:

```
4
3
2
1
0
Key not found
0
1
2
3
4
```

Observation:

The AI correctly identified the missing increment statement and fixed the infinite loop by updating the loop variable inside the loop

Task 6: Task 6: Unpacking Error – Wrong Variables

Tuple unpacking fails because the number of variables does not match the tuple size

Bug: Wrong unpacking

```
a, b = (1, 2, 3)
```

Prompt: # Fix the tuple unpacking error caused by mismatched variables.

Code:

```
#Task 6: Unpacking Error – Wrong Variables
a, b, _ = (1, 2, 3)
print(a, b)
```

Result:

```
3
2
1
0
Key not found
0
1
2
3
4
1 2
```

Observation:

The AI fixed the unpacking issue by using an underscore (_) to ignore extra values, which is a Pythonic and safe practice

Task 7: Task 7: Mixed Indentation – Tabs vs Spaces. Inconsistent indentation causes syntax or runtime errors in Python. # Bug: Mixed indentation

```
def func():
```

```
x = 5
```

```
y = 10
```

```
return x+y
```

Prompt: # Fix the Python code with mixed indentation.

Code:

```
#Task 7: Mixed Indentation – Tabs vs Spaces
def func():
    x = 5
    y = 10
    return x + y

print(func())
```

Result:

```
2
1
0
Key not found
0
1
2
3
4
1 2
15
```

Observation:

The AI resolved the issue by applying consistent indentation using spaces, which is the recommended Python coding standard.

Task 8: Task 8: Import Error – Wrong Module Usage. The code attempts to import a nonexistent module, causing an import error.

Prompt: # Fix the incorrect module import in the Python code.

Bug: Wrong import

```
import maths
```

```
print(maths.sqrt(16))
```

Code:

```
#Task 8: Import Error – Wrong Module Usage
import math
print(math.sqrt(16))
```

Result:

```
1
0
Key not found
0
1
2
3
4
1 2
15
4.0
```

Observation:

The AI correctly identified the incorrect module name and replaced it with the standard `math` module, resolving the import error.